86/563/CDV

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TITLE:

Optical circuit boards - Part 4-214: Interface standards – Terminated waveguide OCB assembly using a single-row thirty-two-channel symmetric PMT connector

PROPOSED STABILITY DATE: 2024

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

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OPTICAL CIRCUIT BOARDS –

Part 4-214: Interface standards – Terminated waveguide OCB assembly using a single-row thirty-two-channel symmetric PMT connector

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International Standard IEC 62496-4-214 has been prepared by IEC technical committee 86: Fibre optics.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
86/XX/FDIS	86/XX/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

OPTICAL CIRCUIT BOARDS –

Part 4-214: Interface standards – Terminated waveguide OCB assembly using a single-row thirty-two-channel symmetric PMT connector

1 Scope

This part of IEC 62496 defines the standard interface dimensions for a terminated waveguide optical circuit board (OCB) assembly (referred to simply as assembly) using single-row thirty-two-channel connectors for polymer waveguides connected with a PMT connector.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 62496-1, *Optical circuit boards – Part 1: General*

IEC 62496-4, *Optical circuit boards – Interface standards – Part4: General and guidance*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 62496-1 and IEC 62496-4 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

IEC Electropedia: available at http://www.electropedia.org/

ISO Online browsing platform: available at http://www.iso.org/obp

4 Description

The assembly is an assembly comprised of a PMT connector and a thirty-two-channel waveguide OCB. The PMT connector is a rectangular connector having the same outer dimensions as the type MT connector specified in IEC 61754-5. The PMT connector is aligned using alignment pins and is normally secured by the use of a latching spring and mates with the type MT connector as shown in Figure 1. Dimensions of components for the assembly are shown in Annex A. The waveguide OCB comprises a planar light-guide consisting of a core and cladding material appropriate to transmit light as the operational wavelengths require, the light-guide being supported on a substrate. Preferably, the substrate will be flexible in order to accommodate compliance to the MT connector. The cores of the waveguide OCB are aligned with the optical fibres of the MT connector after mating using two guide pins and a clamp spring. Dimensions of a single-row thirty-two-channel MT ferrule are shown in Annex B. This PMT connector is not intermateable with the standard 16 channel MT ferrules.

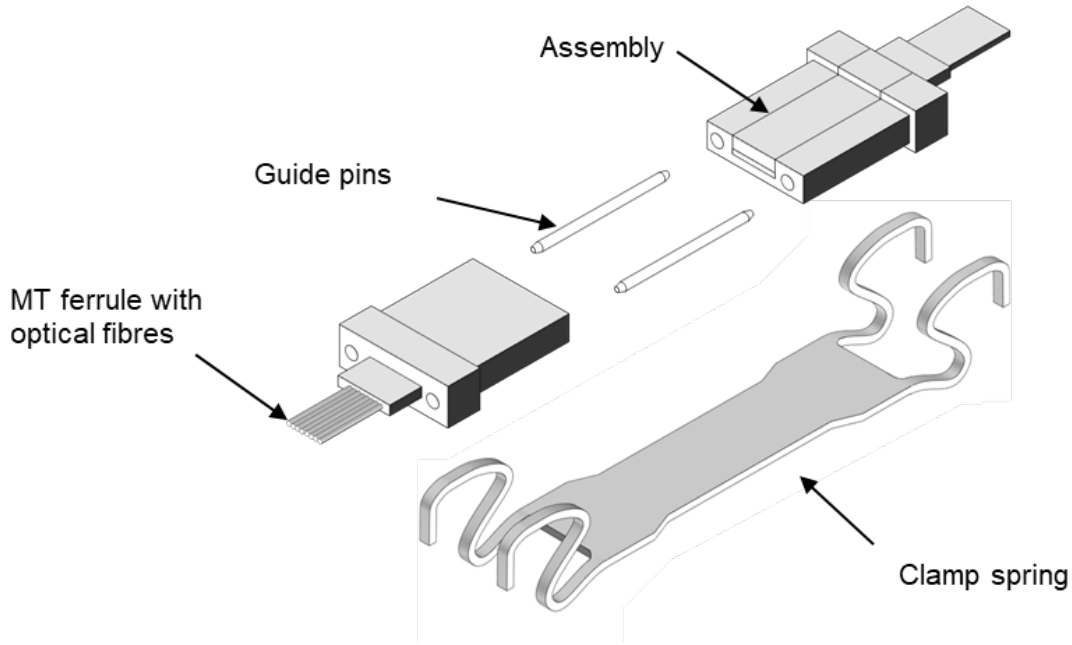
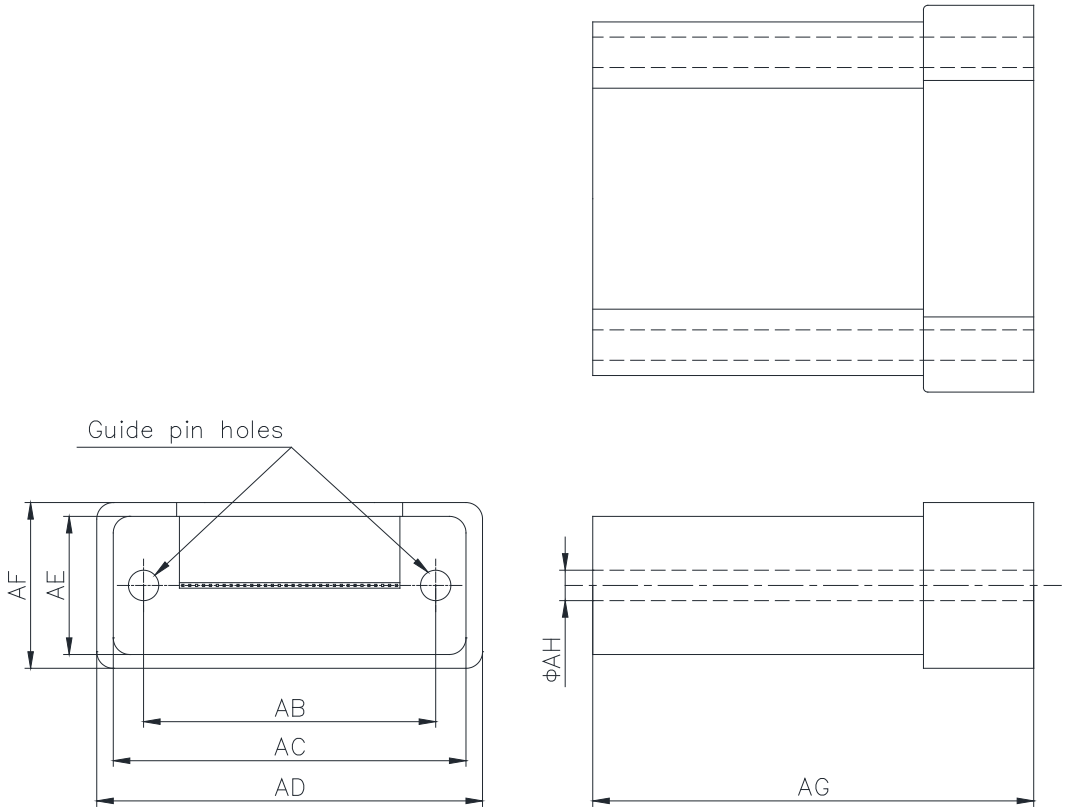


Figure 1 – Interconnection between the assembly and the MT connector

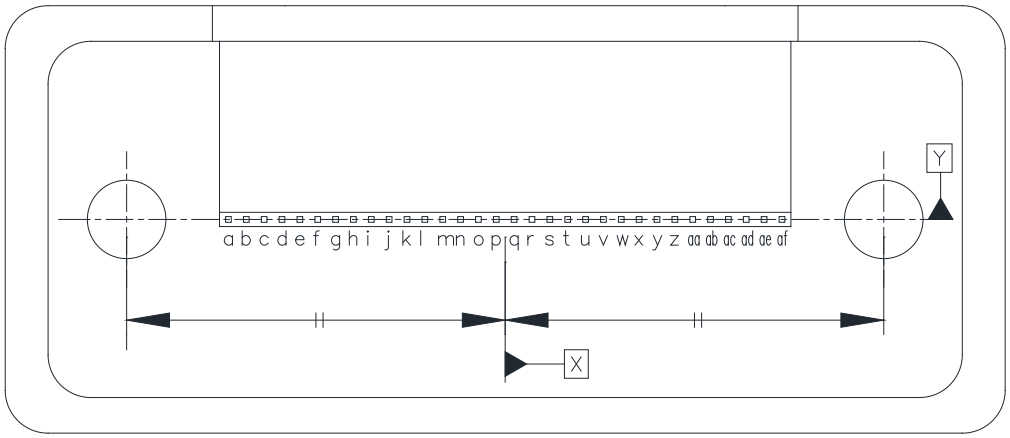
5 Interface dimensions of thirty-two-channel for the assembly, guide pins and clamp spring

Figures 2 and 3 and Table 1 show interface dimensions for the assembly. Table 2 shows positions of the cores for the assembly. The origin point of the assembly is the midpoint of the centres of two guide-holes. Figure 4 and Table 3 show the interface dimensions for a guide pin. Figure 5 and Table 4 show the interface dimensions for a clamp spring.



148

149 **Figure 2 – Interface dimensions of thirty-two-channel for the assembly**



150

151 NOTE The X-axis is defined as the line connecting the two centres of two guide pin holes. The Y axis is defined as
152 the line orthogonal to the X axis intersecting the midpoint of the centres of the two guide pin holes. X shows the
153 zero point of the X-axis and Y shows the zero point of the Y-axis.

154 **Figure 3 – Interface views of thirty-two-channel for the assembly**

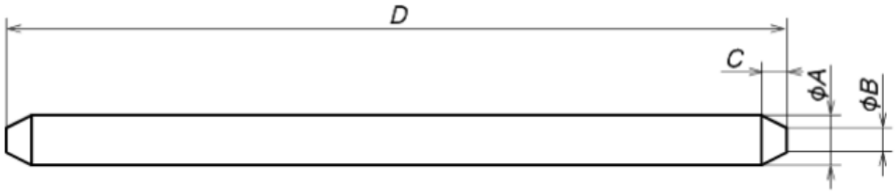


Figure 4 – Interface views of the guide pin

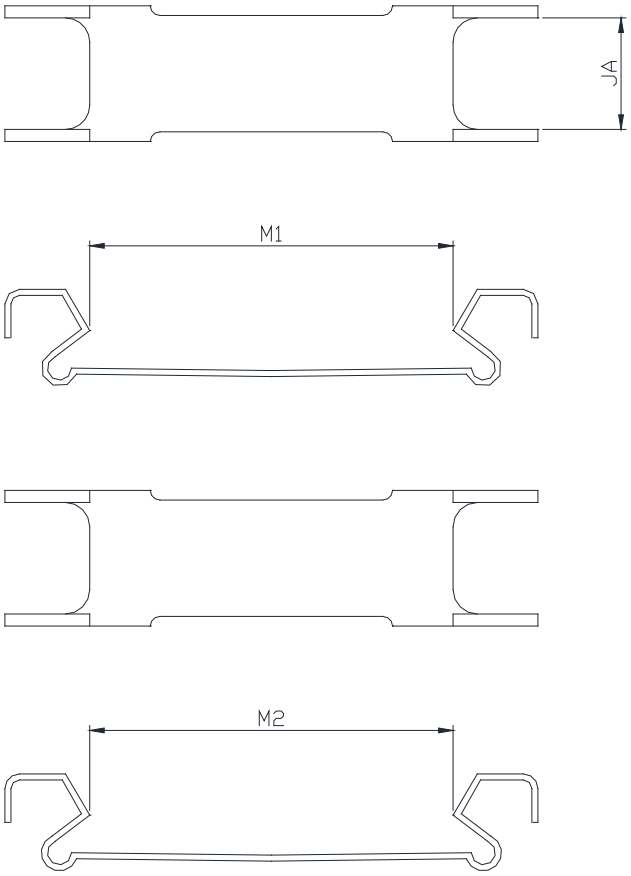


Figure 5 – Interface views of the clamp spring

Table 1 – Interface dimensions of thirty-two-channel for the assembly

Reference	Dimensions mm	
	Minimum	Maximum
AB	5,297	5,303
AC	6,300	6,500
AD	6,900	7,100
AE	2,400	2,500
AF	2,900	3,100
AG	7,900	8,100
AH	0,549	0,551

161 **Table 2 – Positions of cores of thirty-two-channel for the assembly**

Reference	X mm	Y mm	Reference	X mm	Y mm
a	-1,937 5	0,000	q	0,062 5	0,000
b	-1,812 5	0,000	r	0,187 5	0,000
c	-1,687 5	0,000	s	0,312 5	0,000
d	-1,562 5	0,000	t	0,437 5	0,000
e	-1,437 5	0,000	u	0,562 5	0,000
f	-1,312 5	0,000	v	0,687 5	0,000
g	-1,187 5	0,000	w	0,812 5	0,000
h	-1,062 5	0,000	x	0.937 5	0,000
i	-0,937 5	0,000	y	1,062 5	0,000
j	-0,812 5	0,000	z	1,187 5	0,000
k	-0,687 5	0,000	aa	1,312 5	0,000
l	-0,562 5	0,000	ab	1,437 5	0,000
m	-0,437 5	0,000	ac	1,562 5	0,000
n	-0,312 5	0,000	ad	1,687 5	0,000
o	-0,187 5	0,000	ae	1,812 5	0,000
p	-0,062 5	0,000	af	1,937 5	0,000

162

163 **Table 3 – Interface dimensions of the guide pin**

References	Dimensions mm	
	Minimum	Maximum
A	0,547	0,549
B	0,1	0,2
C	0,15	0,35
D	10,8	11,2

164

165 **Table 4 – Interface dimensions of the clamp spring**

References	Dimensions mm	
	Minimum	Maximum
M_1[a]	14,8	15,2
M_2[b]	15,8	16,2
JA	4,55	4,65

[a] Length in free condition.

[b] Compression force for clamping condition shall be 6,8 N to 12,8 N when dimension M_2 is between 15,8 mm and 16,2 mm.

Annex A
(informative)

Dimensions of an example of components for the assembly

A.1 PMT connector

The example of a PMT connector is composed of a PMT body, a PMT cover and a PMT boot, as shown in Figure A.1. The waveguide OCB is embedded and terminated between the PMT body and the PMT cover.

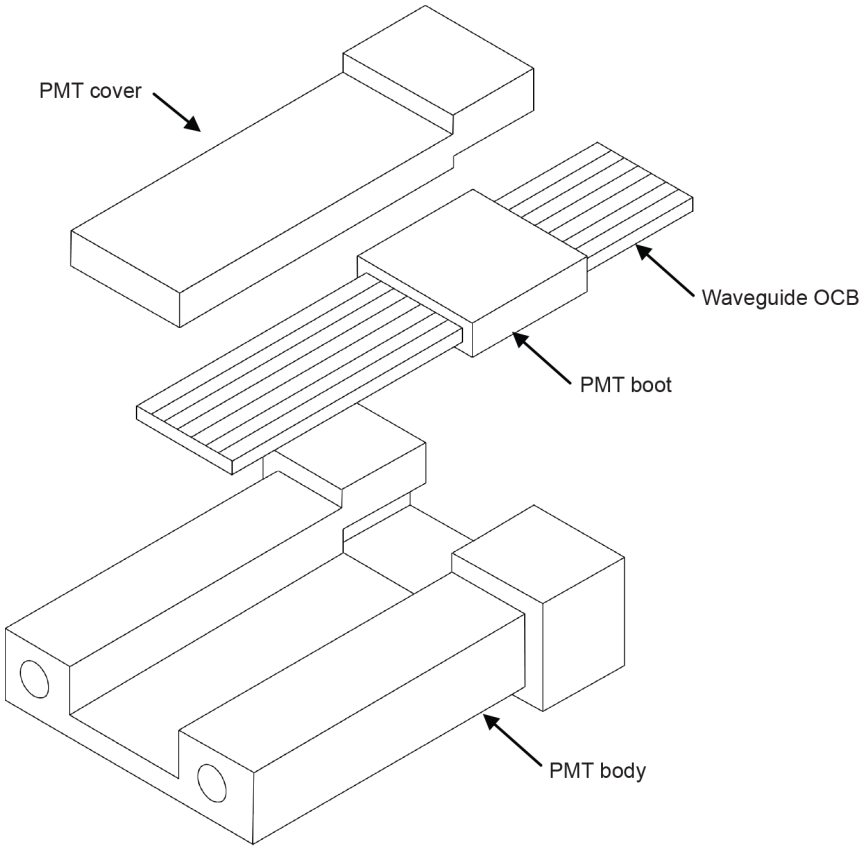
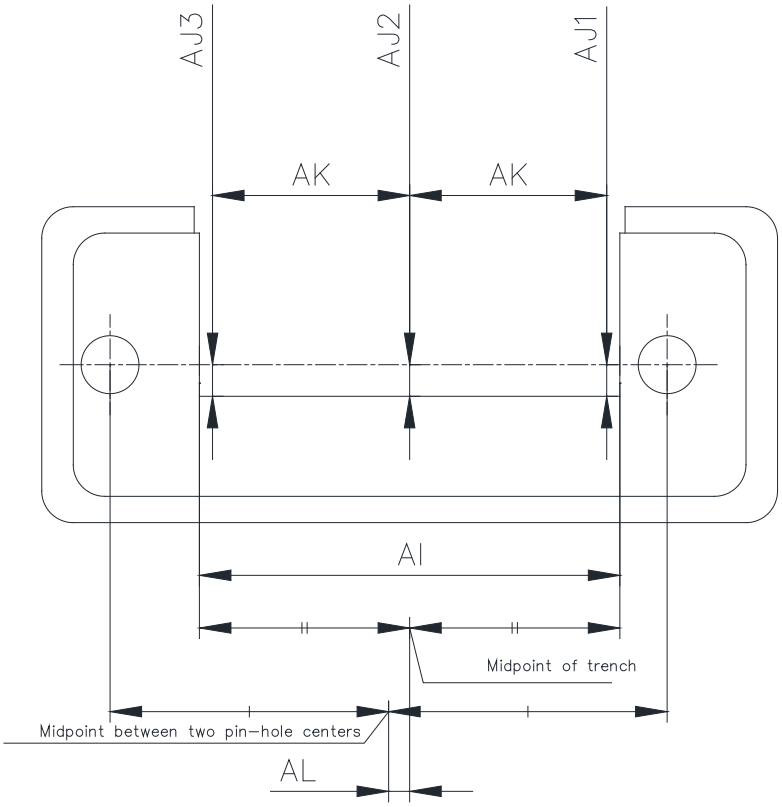


Figure A.1 – An example of components of the PMT connector

Figure A.2 and Table A.1 show the dimensions of an example of the thirty-two-channel PMT body.

178



179

180 **Figure A.2 – Expanded view of end-face for an example of the single-row thirty-two-**
181 **channel PMT body**

182 **Table A.1 – Interface dimensions of an example of the single-row thirty-two-channel for**
183 **PMT body**

Reference	Dimensions mm	
	Minimum	Maximum
AI	4,000	4,003
AJ1	0,047	0,053
AJ2		
AJ3		
AK	1,875	
AL [a]	0	0,003
[a] Gap of midpoint between two pin-hole centres and midpoint of trench.		

184

185 ## A.2 Waveguide OCB

186 Figure A.3 shows the positions of the channel ports at the end-face of an example of the
187 waveguide OCB. The origin point of the waveguide OCB is located at the midpoint of the
188 bottom of the outline of the waveguide OCB, as shown in Figure A.3. Table A.2 gives
189 dimensions of the positions of the example of thirty-two channel ports.

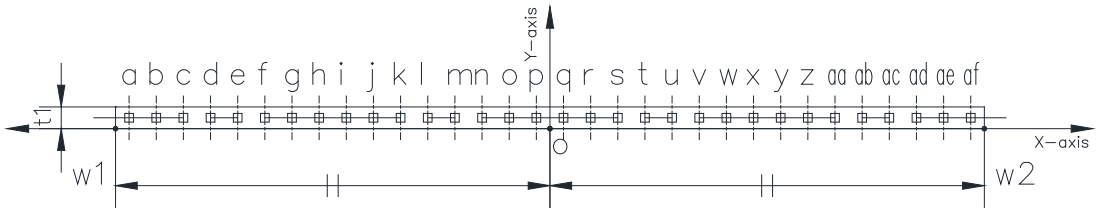


Figure A.3 – Positions of the example of thirty-two channel ports of OCB

Table A.2 – Positions of cores for an example of the thirty-two-channel ports of OCB

Reference	X mm	Y mm	Reference	X mm	Y mm
a	-1,937 5	0,050	q	0,062 5	0,050
b	-1,812 5	0,050	r	0,187 5	0,050
c	-1,687 5	0,050	s	0,312 5	0,050
d	-1,562 5	0,050	t	0,437 5	0,050
e	-1,437 5	0,050	u	0,562 5	0,050
f	-1,312 5	0,050	v	0,687 5	0,050
g	-1,187 5	0,050	w	0,812 5	0,050
h	-1,062 5	0,050	x	0.937 5	0,050
i	-0,937 5	0,050	y	1,062 5	0,050
j	-0,812 5	0,050	z	1,187 5	0,050
k	-0,687 5	0,050	aa	1,312 5	0,050
l	-0,562 5	0,050	ab	1,437 5	0,050
m	-0,437 5	0,050	ac	1,562 5	0,050
n	-0,312 5	0,050	ad	1,687 5	0,050
o	-0,187 5	0,050	ae	1,812 5	0,050
p	-0,062 5	0,050	af	1,937 5	0,050
w1	-2,000	0,050	w2	2,000	0,050
t1		0,100			

Annex B
(informative)

Dimensions of an example of a single-row thirty-two-channel MT ferrule

The MT connector intermateable with a single-row thirty-two-channel PMT connector is not defined yet. Figure B.1 and Table B.1 show an example of a single-row thirty-two-channel MT ferrule used for the customized MT connector. Figure B.2 and Table B.2 show an example of the interface view and fibre hole positions, respectively. Fibre types for the customized MT connector are those with a cladding, for example, of Φ80 μm and a coating of Φ125 μm.

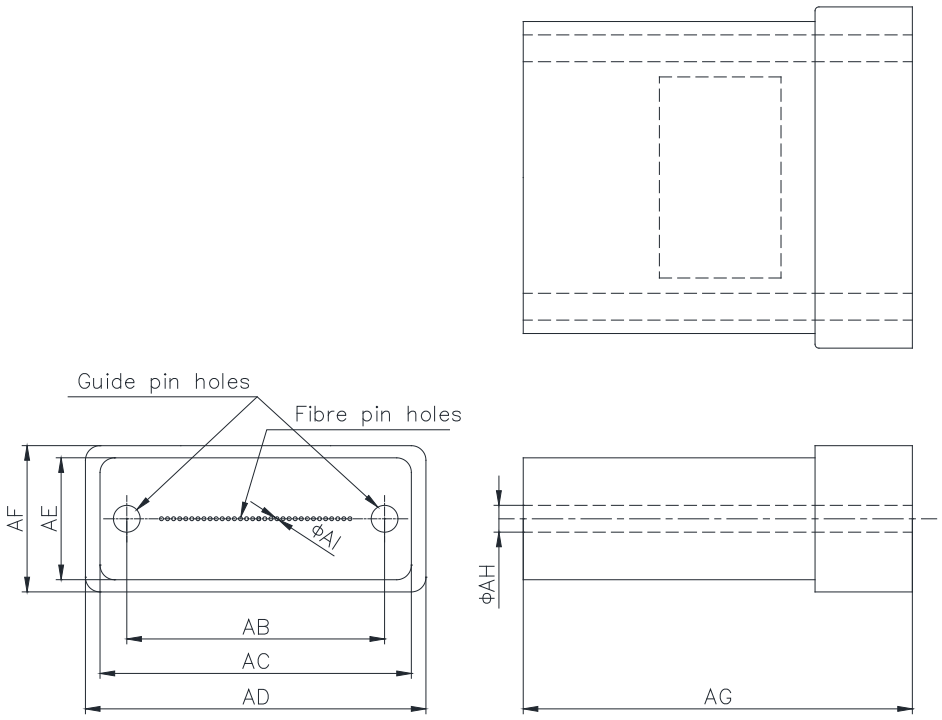


Figure B.1 – Interface dimensions of an example of single-row thirty-two-channel MT ferrule

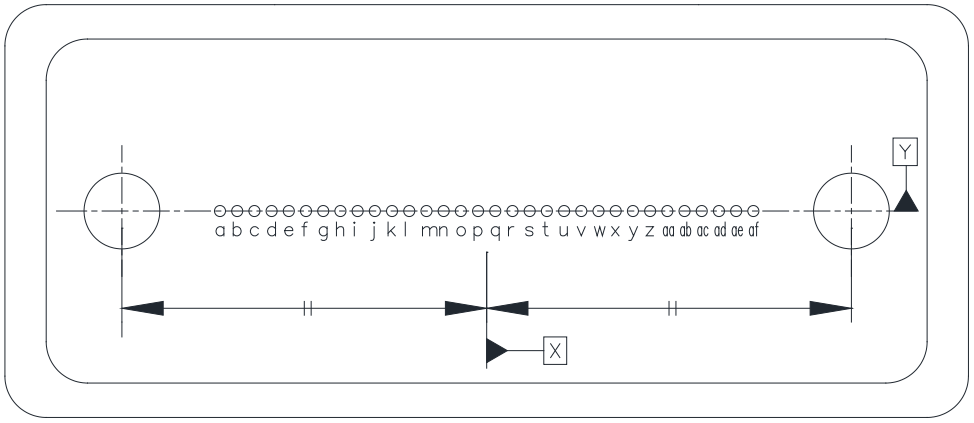


Figure B.2 – Interface views of an example of single-row thirty-two-channel MT ferrule

209 **Table B.1 – Interface dimensions of an example of single-row thirty-two-channel MT**
210 **ferrule**

Reference	Dimensions mm	
	Minimum	Maximum
AB	5,297	5,303
AC	6,300	6,500
AD	6,900	7,100
AE	2,400	2,500
AF	2,900	3,100
AG	7,900	8,100
AH	0,549	0,551
AI	0,080 0	0,081 5

211

212 **Table B.2 – Fibre hole positions of an example of single-row thirty-two-channel MT**
213 **ferrule**

Reference	X mm	Y mm	Reference	X mm	Y mm
a	-1,937 5	0,000	q	0,062 5	0,000
b	-1,812 5	0,000	r	0,187 5	0,000
c	-1,687 5	0,000	s	0,312 5	0,000
d	-1,562 5	0,000	t	0,437 5	0,000
e	-1,437 5	0,000	u	0,562 5	0,000
f	-1,312 5	0,000	v	0,687 5	0,000
g	-1,187 5	0,000	w	0,812 5	0,000
h	-1,062 5	0,000	x	0.937 5	0,000
i	-0,937 5	0,000	y	1,062 5	0,000
j	-0,812 5	0,000	z	1,187 5	0,000
k	-0,687 5	0,000	aa	1,312 5	0,000
l	-0,562 5	0,000	ab	1,437 5	0,000
m	-0,437 5	0,000	ac	1,562 5	0,000
n	-0,312 5	0,000	ad	1,687 5	0,000
o	-0,187 5	0,000	ae	1,812 5	0,000
p	-0,062 5	0,000	af	1,937 5	0,000

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215 # Bibliography

216 IEC 61754-5, *Fibre optic connector interfaces - Part 5: Type MT connector family*

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